

Natural User Interface: Empowerment Beyond Touch and Feel

A key goal for natural user interface is to pave the way for the devices and systems that will empower the ones with limitations. Developers aim at making the latest technology more accessible to those with disabilities like physical impairments or learning difficulties



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It has long been in the realm of science fiction that one can control and have absolute mastery over everything around simply with a wave of hand or a voice command. With recent innovations in motion-sensing technology bringing this into our very living rooms, it is time to look closely at how technology is poised to revolutionise the way we perceive computing.

Motion-sensing devices offer natural interactive experiences. Gamers are able to turn into sport stars, racing car drivers and action heroes simply by moving their limbs and/or jumping around their play-space. All this has been made possible with the development of 'gesture recognition technology'.

Gesture recognition technology involves a camera that reads the movements of the human body and communicates that data to a PC that uses the gestures as inputs to control devices or applications. Motion-sensing technology and depth perception are used to accurately interpret human movements and enhance user experience.

Gesture recognition is just the beginning of the movement towards natural user interface (NUI), which excites innovators across the spectrum of technology. NUI marks one of the most significant ongoing trends in digital technology and has the poten-

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tial to transform future human-device interactions. In effect, it is an evolution from current exciting features like touchscreen, multi-touch functions and voice recognition. This evolution is coming about as the technology is going through a crucial shift—computing platforms are evolving; there are huge changes in microprocessor architecture, which enable new opportunities; and a new set of applications where new user interface experiences will emerge and contribute significantly in areas like energy, education and healthcare.

These developments are exciting not only for those who work in

the technology industry but also for people across the globe who adopt these advancements with enthusiasm. Gesture-based gaming has generated great interest among users. With the changing and growing desire for technology, it is important to understand changing trends in user needs. Individuals are steadily moving from just using technology to simplify lives to using it for recreation. There is a gradual shift by people towards exploring different technologies for broadening the level of technology integration in their own lives.

As with other technological developments, a key goal for NUI is to pave the way for the devices and systems which will empower the ones with limitations. Developers aim at making the latest technology more accessible to those with disabilities like physical impairments or learning difficulties. At the *Pacific Health Summit* in June this year, Microsoft demonstrated how medical providers can use technology to improve healthcare facilities. In the US, therapists working with autistic children discovered that integrating Kinect's full-body play technology into their therapy sessions had a positive effect on kids' engagement. It is just the beginning and the potential is huge.

The future is to take computers from just being tools to becoming helpers with a certain degree of autonomy to help understand you and me better, and become more intuitive. ●